VII) CURRICULUM STRUCTURE

SEMESTER – I

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1	20HSG01	Technical English	HS	2	0	2	4	3
2	20MHG01	Calculus and Linear Algebra	BS	3	1	0	4	4
3	20CHG01	Engineering Chemistry	BS	3	1	0	4	4
4	20ITG01	Programming for Problem Solving using C	ES	3	0	0	3	3
5	20EEG01	Basic Electrical and Electronics Engineering	ES	3	0	0	3	3
PRACTICALS								
6	20MEG01	Engineering Graphics	ES	0	0	4	4	2
7	20CHG02	Engineering Chemistry Laboratory	BS	0	0	3	3	1.5
8	20ITG02	Programming in C Laboratory	ES	0	0	4	4	2
			Total	**14**	**2**	**13**	**29**	**22.5**

SEMESTER – II

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1	20HSG02	Universal Human Values – II Understanding Harmony	HS	3	0	0	3	3
2	20MHG02	Differential Equations and Complex Variables	BS	3	1	0	4	4
3	20PHG01	Engineering Physics	BS	3	1	0	4	4
4	20CSG01	Object Oriented Programming using C++	ES	3	0	0	3	3
5	20HMG01	Information Technology Essentials and Ethics	HM	3	0	0	3	3
6	20AC001	Environmental Science and Engineering	AC	3	0	0	3	0
PRACTICALS								
7	20MEG02	Engineering Workshop	ES	0	0	4	4	2
8	20PHG02	Engineering Physics Laboratory	BS	0	0	3	3	1.5
9	20CSG02	Programming in C++ Laboratory	ES	0	0	4	4	2
			Total	**18**	**2**	**11**	**31**	**22.5**

SEMESTER – III

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1	20MHG06	Probability and Statistics	BS	3	1	0	4	4
2	20IT001	Data Structures	PCC	3	0	0	3	3
3	20IT002	Java Programming	PCC	3	0	0	3	3
4	20CS001	Computer Organization and Architecture	PCC	3	0	0	3	3
5	20ECG02	Digital Principles and System Design	ES	3	0	0	3	3
6	20XXEXX	Open Elective - I	OE	3	0	0	3	3
PRACTICALS								
7	20IT003	Data Structures Laboratory	PCC	0	0	3	3	1.5
8	20IT004	Java Programming Laboratory	PCC	0	0	3	3	1.5
			Total	**18**	**1**	**6**	**25**	**22**

SEMESTER – IV

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1	20IT005	Discrete Structures	PCC	3	0	0	3	3
2	20IT006	Computer Networks	PCC	3	0	0	3	3
3	20CS004	Software Engineering	PCC	3	0	0	3	3
4	20IT007	Database	PCC	3	0	0	3	3

		Management Systems						
5	20IT008	Operating Systems	PCC	3	0	0	3	3
6	20XXEXX	Open Elective - II	OE	3	0	0	3	3
PRACTICALS								
7	20IT009	Database Management Systems Laboratory	PCC	0	0	3	3	1.5
8	20IT010	Operating Systems Laboratory	PCC	0	0	3	3	1.5
			Total	**18**	**0**	**6**	**24**	**21**

SEMESTER – V

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1	20IT011	Compiler Design	PCC	3	0	0	3	3
2	20IT012	Design and Analysis of Algorithms	PCC	3	0	0	3	3
3	20CS007	Object Oriented Analysis and Design	PCC	3	0	0	3	3
4	20IT013	Web Technology	PCC	3	0	0	3	3
5	20ITPXX	Professional Elective - I	PE	3	0	0	3	3
6	20XXEXX	Open Elective - III	OE	3	0	0	3	3
PRACTICALS								
7	20CS011	CASE Tools Laboratory	PCC	0	0	3	3	1.5
8	20IT014	Web Technology Laboratory	PCC	0	0	3	3	1.5
			Total	**18**	**0**	**6**	**24**	**21**

SEMESTER – VI

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1	20IT015	Machine Learning	PCC	3	0	0	3	3
2	20IT016	Internet of Things	PCC	3	0	0	3	3
3	20IT017	Data Analytics	PCC	3	0	0	3	3
4	20IT018	Cloud Computing	PCC	3	0	0	3	3
5	20ITPXX	Professional Elective - II	PE	3	0	0	3	3
6	20AC002	Constitution of India	AC	3	0	0	3	0
PRACTICALS								
7	20IT019	Data Analytics and Cloud Laboratory	PCC	0	0	3	3	1.5
8	20IT020	Internet of Things Laboratory	PCC	0	0	3	3	1.5
9	20IT901	Design Project	EC	0	0	8	8	4
			Total	**18**	**0**	**14**	**32**	**22**

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SEMESTER – VII

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1	20IT021	Cryptography and Network Security	PCC	3	0	0	3	3
2	20ITPXX	Professional Elective - III	PE	3	0	0	3	3
3	20ITPXX	Professional Elective - IV	PE	3	0	0	3	3
4	20ITPXX	Professional Elective - V	PE	3	0	0	3	3
5	20XXEXX	Open Elective - IV	OE	3	0	0	3	3
PRACTICALS								
6	20IT902	Project Phase - I	EC	0	0	6	6	3
			Total	**15**	**0**	**6**	**21**	**18**

SEMESTER – VIII

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1	20ITPXX	Professional Elective - VI	PE	3	0	0	3	3
PRACTICALS								
2	20IT903	Final Year Project	EC	0	0	16	16	8
			Total	**3**	**0**	**16**	**19**	**11**

TOTAL NUMBER OF CREDITS: 160

PROFESSIONAL ELECTIVES

ELECTIVE – I [PROGRAMMING]

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1	20ITP11	Python Programming	PE	3	0	0	3	3
2	20ITP12	Advanced Java Programming	PE	3	0	0	3	3
3	20ITP13	Unix Internals	PE	3	0	0	3	3
4	20ITP14	C# and .Net Framework	PE	3	0	0	3	3
5	20ITP15	Programming with Open source Software	PE	3	0	0	3	3
6	20ITP16	Translators and System Software	PE	3	0	0	3	3
7	20ITP17	Introduction to Artificial Intelligence	PE	3	0	0	3	3

ELECTIVE – II [NETWORKING]

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1	20ITP21	Distributed Systems	PE	3	0	0	3	3
2	20ITP22	Mobile Computing	PE	3	0	0	3	3
3	20ITP23	Communication Switching Techniques	PE	3	0	0	3	3
4	20ITP24	Ad-hoc and Sensor	PE	3	0	0	3	3

		Networks						
5	20ITP25	Software Defined Networks	PE	3	0	0	3	3
6	20ITP26	Quantum Computing	PE	3	0	0	3	3
7	20ITP42	Data Mining and Data Warehousing	PE	3	0	0	3	3

ELECTIVE – III [HUMAN COMPUTER INTERACTION]

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1	20ITP31	Digital Image Processing	PE	3	0	0	3	3
2	20ITP32	Pattern Recognition	PE	3	0	0	3	3
3	20ITP33	User Experience Design	PE	3	0	0	3	3
4	20ITP34	Computer Graphics and Multimedia	PE	3	0	0	3	3
5	20ITP35	Game Programming	PE	3	0	0	3	3
6	20ITP36	Introduction to Augmented Reality	PE	3	0	0	3	3

ELECTIVE – IV [DATABASES]

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1	20ITP41	Advanced Database Technology	PE	3	0	0	3	3
2	20ITP43	Predictive Analytics	PE	3	0	0	3	3
3	20ITP44	Deep Learning	PE	3	0	0	3	3

4	20ITP45	Parallel Computing	PE	3	0	0	3	3
5	20ITP46	Business Intelligence	PE	3	0	0	3	3

ELECTIVE – V [WEB SYSTEMS]

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1	20ITP51	Web Engineering	PE	3	0	0	3	3
2	20ITP52	Advanced Web Programming	PE	3	0	0	3	3
3	20ITP53	PHP and MySQL	PE	3	0	0	3	3
4	20ITP54	Mobile Application Development	PE	3	0	0	3	3
5	20ITP55	Service Oriented Architecture	PE	3	0	0	3	3
6	20ITP56	Blockchains	PE	3	0	0	3	3

ELECTIVE – VI [INFORMATION ASSURANCE AND SECURITY]

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1	20ITP61	Management Information Systems	PE	3	0	0	3	3
2	20ITP62	Software Testing	PE	3	0	0	3	3
3	20ITP63	Software Quality Assurance	PE	3	0	0	3	3
4	20ITP64	Principles of Cyber Security	PE	3	0	0	3	3
5	20ITP65	IT Security and Risk Management	PE	3	0	0	3	3

6	20ITP66	Secure Software Development	PE	3	0	0	3	3

OPEN ELECTIVES

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
OFFERED BY ELECTRICAL AND ELECTRONICS ENGINEERING PROGRAMME								
1	20EEE01	Energy Management Systems	OE	3	0	0	3	3
2	20EEE02	Medical Instrumentation	OE	3	0	0	3	3
3	20EEE03	PLC Programming	OE	3	0	0	3	3
4	20EEE04	Renewable Energy Systems	OE	3	0	0	3	3
5	20EEE05	Virtual Instrumentation and Data Acquisition	OE	3	0	0	3	3
6	20EEE06	Electric Vehicles	OE	3	0	0	3	3
OFFERED BY ELECTRONICS AND COMMUNICATION ENGINEERING PROGRAMME								
1	20ECE01	Electronic Measurements and Instrumentation	OE	3	0	0	3	3
2	20ECE02	Microcontrollers and its Applications	OE	3	0	0	3	3
3	20ECE03	Introduction to Embedded Systems	OE	3	0	0	3	3

4	20ECE04	Nano Electronics and Sensors	OE	3	0	0	3	3
5	20ECE05	Principles of VLSI Systems	OE	3	0	0	3	3
OFFERED BY MECHNAICAL ENGINEERING PROGRAMME								
1	20MEE01	Automotive Fundamentals	OE	3	0	0	3	3
2	20MEE02	Computer Aided Design	OE	3	0	0	3	3
3	20MEE03	Power Plant Engineering	OE	3	0	0	3	3
4	20MEE04	Industrial Engineering	OE	3	0	0	3	3
5	20MEE05	Rapid Prototyping	OE	3	0	0	3	3

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